OHIDE ANDREW ODOHU.
DATABASE WEEK6 ASSIGNMENT

```
# -----------------------------
# PostgreSQL configuration file
# -----------------------------
#
# This file consists of lines of the form:
#
#   name = value
#
# (The "=" is optional.)  Whitespace may be used.  Comments are introduced with
# "#" anywhere on a line.  The complete list of parameter names and allowed
# values can be found in the PostgreSQL documentation.
#
# The commented-out settings shown in this file represent the default values.
# Re-commenting a setting is NOT sufficient to revert it to the default value;
# you need to reload the server.
#
# This file is read on server startup and when the server receives a SIGHUP
# signal.  If you edit the file on a running system, you have to SIGHUP the
# server for the changes to take effect, run "pg_ctl reload", or execute
# "SELECT pg_reload_conf()".  Some parameters, which are marked below,
# require a server shutdown and restart to take effect.
#
# Any parameter can also be given as a command-line option to the server, e.g.,
# "postgres -c log_connections=all".  Some parameters can be changed at run time
# with the "SET" SQL command.
#
# Memory units:  B  = bytes            Time units:  us  = microseconds
#                kB = kilobytes                ms  = milliseconds
#                MB = megabytes                s   = seconds
#                GB = gigabytes                min = minutes
#                TB = terabytes                h   = hours
#                                              d   = days


#------------------------------------------------------------------------------
# FILE LOCATIONS
#------------------------------------------------------------------------------

# The default values of these variables are driven from the -D command-line
# option or PGDATA environment variable, represented here as ConfigDir.

#data_directory = 'ConfigDir'           # use data in another directory
                                        # (change requires restart)
#hba_file = 'ConfigDir/pg_hba.conf'     # host-based authentication file
```

```
                                        # (change requires restart)
#ident_file = 'ConfigDir/pg_ident.conf'         # ident configuration file
                                        # (change requires restart)

# If external_pid_file is not explicitly set, no extra PID file is written.
#external_pid_file = ''                 # write an extra PID file
                                        # (change requires restart)


#------------------------------------------------------------------------------
# CONNECTIONS AND AUTHENTICATION
#------------------------------------------------------------------------------

# - Connection Settings -

listen_addresses = '*'
                                        # comma-separated list of addresses;
                                        # defaults to 'localhost'; use '*' for all
                                        # (change requires restart)
port = 5432
max_connections = 100                   # (change requires restart)
#reserved_connections = 0               # (change requires restart)
#superuser_reserved_connections = 3     # (change requires restart)
#unix_socket_directories = ''           # comma-separated list of directories
                                        # (change requires restart)
#unix_socket_group = ''                 # (change requires restart)
#unix_socket_permissions = 0777         # begin with 0 to use octal notation
                                        # (change requires restart)
#bonjour = off                          # advertise server via Bonjour
                                        # (change requires restart)
#bonjour_name = ''                      # defaults to the computer name
                                        # (change requires restart)

# - TCP settings -
# see "man tcp" for details

#tcp_keepalives_idle = 0                # TCP_KEEPIDLE, in seconds;
                                        # 0 selects the system default
#tcp_keepalives_interval = 0            # TCP_KEEPINTVL, in seconds;
                                        # 0 selects the system default
#tcp_keepalives_count = 0               # TCP_KEEPCNT;
                                        # 0 selects the system default
#tcp_user_timeout = 0                   # TCP_USER_TIMEOUT, in milliseconds;
                                        # 0 selects the system default

#client_connection_check_interval = 0   # time between checks for client
                                        # disconnection while running queries;
                                        # 0 for never
```

```
# - Authentication -

#authentication_timeout = 1min                  # 1s-600s
#password_encryption = scram-sha-256      # scram-sha-256 or (deprecated) md5
#scram_iterations = 4096
#md5_password_warnings = on                  # display md5 deprecation warnings?
#oauth_validator_libraries = ''# comma-separated list of trusted validator modules

# GSSAPI using Kerberos
#krb_server_keyfile = 'FILE:${sysconfdir}/krb5.keytab'
#krb_caseins_users = off
#gss_accept_delegation = off

# - SSL -

#ssl = off
#ssl_ca_file = ''
#ssl_cert_file = 'server.crt'
#ssl_crl_file = ''
#ssl_crl_dir = ''
#ssl_key_file = 'server.key'
#ssl_ciphers = 'HIGH:MEDIUM:+3DES:!aNULL'      # allowed TLSv1.2 ciphers
#ssl_tls13_ciphers = ''# allowed TLSv1.3 cipher suites, blank for default
#ssl_prefer_server_ciphers = on
#ssl_groups = 'X25519:prime256v1'
#ssl_min_protocol_version = 'TLSv1.2'
#ssl_max_protocol_version = ''
#ssl_dh_params_file = ''
#ssl_passphrase_command = ''
#ssl_passphrase_command_supports_reload = off


#------------------------------------------------------------------------------
# RESOURCE USAGE (except WAL)
#------------------------------------------------------------------------------

# - Memory -

shared_buffers = 128MB                  # min 128kB
                                        # (change requires restart)
#huge_pages = try                       # on, off, or try
                                        # (change requires restart)
#huge_page_size = 0                     # zero for system default
                                        # (change requires restart)
#temp_buffers = 8MB                     # min 800kB
#max_prepared_transactions = 0          # zero disables the feature
                                        # (change requires restart)
```

```
# Caution: it is not advisable to set max_prepared_transactions nonzero unless
# you actively intend to use prepared transactions.
#work_mem = 4MB                         # min 64kB
#hash_mem_multiplier = 2.0              # 1-1000.0 multiplier on hash table work_mem
#maintenance_work_mem = 64MB            # min 64kB
#autovacuum_work_mem = -1               # min 64kB, or -1 to use
maintenance_work_mem
#logical_decoding_work_mem = 64MB       # min 64kB
#max_stack_depth = 2MB                  # min 100kB
#shared_memory_type = mmap              # the default is the first option
                                        # supported by the operating system:
                                        #   mmap
                                        #   sysv
                                        #   windows
                                        # (change requires restart)
dynamic_shared_memory_type = windows  # the default is usually the first option
                                        # supported by the operating system:
                                        #   posix
                                        #   sysv
                                        #   windows
                                        #   mmap
                                        # (change requires restart)
#min_dynamic_shared_memory = 0MB       # (change requires restart)
#vacuum_buffer_usage_limit = 2MB # size of vacuum and analyze buffer access strategy
ring;
                                        # 0 to disable vacuum buffer access strategy;
                                        # range 128kB to 16GB

# SLRU buffers (change requires restart)
#commit_timestamp_buffers = 0           # memory for pg_commit_ts (0 = auto)
#multixact_offset_buffers = 16          # memory for pg_multixact/offsets
#multixact_member_buffers = 32          # memory for pg_multixact/members
#notify_buffers = 16                    # memory for pg_notify
#serializable_buffers = 32              # memory for pg_serial
#subtransaction_buffers = 0             # memory for pg_subtrans (0 = auto)
#transaction_buffers = 0                # memory for pg_xact (0 = auto)

# - Disk -

#temp_file_limit = -1                   # limits per-process temp file space
                                        # in kilobytes, or -1 for no limit

#file_copy_method = copy                # copy, clone (if supported by OS)
#file_extend_method = posix_fallocate   # the default is the first option supported
                                        # by the operating system:
                                        #   posix_fallocate (most Unix-like systems)
                                        #   write_zeros
```

```
#max_notify_queue_pages = 1048576       # limits the number of SLRU pages allocated
                                        # for NOTIFY / LISTEN queue

# - Kernel Resources -

#max_files_per_process = 1000           # min 64
                                        # (change requires restart)

# - Background Writer -

#bgwriter_delay = 200ms                 # 10-10000ms between rounds
#bgwriter_lru_maxpages = 100            # max buffers written/round, 0 disables
#bgwriter_lru_multiplier = 2.0          # 0-10.0 multiplier on buffers scanned/round
#bgwriter_flush_after = 0               # measured in pages, 0 disables

# - I/O -

#backend_flush_after = 0                # measured in pages, 0 disables
#effective_io_concurrency = 16          # 1-1000; 0 disables issuing multiple
simultaneous IO requests
#maintenance_io_concurrency = 16        # 1-1000; same as effective_io_concurrency
#io_max_combine_limit = 128kB           # usually 1-128 blocks (depends on OS)
                                        # (change requires restart)
#io_combine_limit = 128kB               # usually 1-128 blocks (depends on OS)

#io_method = worker                     # worker, io_uring, sync
                                        # (change requires restart)
#io_max_concurrency = -1                # Max number of IOs that one process
                                        # can execute simultaneously
                                        # -1 sets based on shared_buffers
                                        # (change requires restart)
#io_workers = 3                         # 1-32;

# - Worker Processes -

#max_worker_processes = 8               # (change requires restart)
#max_parallel_workers_per_gather = 2    # limited by max_parallel_workers
#max_parallel_maintenance_workers = 2   # limited by max_parallel_workers
#max_parallel_workers = 8               # number of max_worker_processes that
                                        # can be used in parallel operations
#parallel_leader_participation = on


#------------------------------------------------------------------------------
# WRITE-AHEAD LOG
#------------------------------------------------------------------------------

# - Settings -
```

```
#wal_level = replica = replica                  # minimal, replica, or logical
                                        # (change requires restart)
#fsync = on                             # flush data to disk for crash safety
                                        # (turning this off can cause
                                        # unrecoverable data corruption)
#synchronous_commit = on                # synchronization level;
                                        # off, local, remote_write, remote_apply, or on
#wal_sync_method = fsync                # the default is the first option
                                        # supported by the operating system:
                                        #   open_datasync
                                        #   fdatasync (default on Linux and FreeBSD)
                                        #   fsync
                                        #   fsync_writethrough
                                        #   open_sync
#full_page_writes = on                          # recover from partial page writes
#wal_log_hints = off                    # also do full page writes of non-critical updates
                                        # (change requires restart)
#wal_compression = off                          # enables compression of full-page writes;
                                        # off, pglz (or "on"), lz4, or zstd
#wal_init_zero = on                     # zero-fill new WAL files
#wal_recycle = on                       # recycle WAL files
#wal_buffers = -1                       # min 32kB, -1 sets based on shared_buffers
                                        # (change requires restart)
#wal_writer_delay = 200ms               # 1-10000 milliseconds
#wal_writer_flush_after = 1MB                   # measured in pages, 0 disables
#wal_skip_threshold = 2MB

#commit_delay = 0                       # range 0-100000, in microseconds
#commit_siblings = 5                    # range 0-1000

# - Checkpoints -

#checkpoint_timeout = 5min              # range 30s-1d
#checkpoint_completion_target = 0.9# checkpoint target duration, 0.0 - 1.0
#checkpoint_flush_after = 0             # measured in pages, 0 disables
#checkpoint_warning = 30s               # 0 disables
max_wal_size = 1GB
min_wal_size = 80MB

# - Prefetching during recovery -

#recovery_prefetch = try        # prefetch pages referenced in the WAL?
#wal_decode_buffer_size = 512kB    # lookahead window used for prefetching
                                # (change requires restart)

# - Archiving -
```

```
#archive_mode = on             # enables archiving; off, on, or always
                               # (change requires restart)
#archive_library = ''          # library to use to archive a WAL file
                               # (empty string indicates archive_command = 'copy /Y "%p"
"C:/pgbackup/wal/%f"' should
                               # be used)
#archive_command = ''                  # command to use to archive a WAL file
                               # placeholders: %p = path of file to archive
                               #               %f = file name only
                               # e.g. 'test ! -f /mnt/server/archivedir/%f && cp %p
/mnt/server/archivedir/%f'
#archive_timeout = 0           # force a WAL file switch after this
                               # number of seconds; 0 disables

# - Archive Recovery -

# These are only used in recovery mode.

#restore_command = ''                  # command to use to restore an archived WAL file
                               # placeholders: %p = path of file to restore
                               #               %f = file name only
                               # e.g. 'cp /mnt/server/archivedir/%f %p'
#archive_cleanup_command = ''       # command to execute at every restartpoint
#recovery_end_command = ''# command to execute at completion of recovery

# - Recovery Target -

# Set these only when performing a targeted recovery.

#recovery_target = ''          # 'immediate' to end recovery as soon as a
                               # consistent state is reached
                               # (change requires restart)
#recovery_target_name = ''    # the named restore point to which recovery will proceed
                               # (change requires restart)
#recovery_target_time = ''     # the time stamp up to which recovery will proceed
                               # (change requires restart)
#recovery_target_xid = ''      # the transaction ID up to which recovery will proceed
                               # (change requires restart)
#recovery_target_lsn = ''      # the WAL LSN up to which recovery will proceed
                               # (change requires restart)
#recovery_target_inclusive = on       # Specifies whether to stop:
                               # just after the specified recovery target (on)
                               # just before the recovery target (off)
                               # (change requires restart)
#recovery_target_timeline = 'latest'   # 'current', 'latest', or timeline ID
                                       # (change requires restart)
#recovery_target_action = 'pause'     # 'pause', 'promote', 'shutdown'
                                       # (change requires restart)
```

```
# - WAL Summarization -

#summarize_wal = off           # run WAL summarizer process?
#wal_summary_keep_time = '10d'     # when to remove old summary files, 0 = never


#------------------------------------------------------------------------------
# REPLICATION
#------------------------------------------------------------------------------

# - Sending Servers -

# Set these on the primary and on any standby that will send replication data.

#max_wal_senders = 10                   # max number of walsender processes
                                # (change requires restart)
#max_replication_slots = 10  # max number of replication slots
                                # (change requires restart)
#wal_keep_size = 0              # in megabytes; 0 disables
#max_slot_wal_keep_size = -1        # in megabytes; -1 disables
#idle_replication_slot_timeout = 0      # in seconds; 0 disables
#wal_sender_timeout = 60s   # in milliseconds; 0 disables
#track_commit_timestamp = off       # collect timestamp of transaction commit
                                # (change requires restart)

# - Primary Server -

# These settings are ignored on a standby server.

#synchronous_standby_names = ''    # standby servers that provide sync rep
                                # method to choose sync standbys, number of sync standbys,
                                # and comma-separated list of application_name
                                # from standby(s); '*' = all
#synchronized_standby_slots = ''       # streaming replication standby server slot
                                # names that logical walsender processes will wait for

# - Standby Servers -

# These settings are ignored on a primary server.

#primary_conninfo = ''                  # connection string to sending server
#primary_slot_name = ''                         # replication slot on sending server
#hot_standby = on                       # "off" disallows queries during recovery
                                # (change requires restart)
#max_standby_archive_delay = 30s # max delay before canceling queries
                                # when reading WAL from archive;
                                # -1 allows indefinite delay
```

```
#max_standby_streaming_delay = 30s      # max delay before canceling queries
                                        # when reading streaming WAL;
                                        # -1 allows indefinite delay
#wal_receiver_create_temp_slot = off    # create temp slot if primary_slot_name
                                        # is not set
#wal_receiver_status_interval = 10s     # send replies at least this often
                                        # 0 disables
#hot_standby_feedback = off             # send info from standby to prevent
                                        # query conflicts
#wal_receiver_timeout = 60s             # time that receiver waits for
                                        # communication from primary
                                        # in milliseconds; 0 disables
#wal_retrieve_retry_interval = 5s       # time to wait before retrying to
                                        # retrieve WAL after a failed attempt
#recovery_min_apply_delay = 0           # minimum delay for applying changes during
recovery
#sync_replication_slots = off           # enables slot synchronization on the physical standby
from the primary

# - Subscribers -

# These settings are ignored on a publisher.

#max_active_replication_origins = 10    # max number of active replication origins
                                        # (change requires restart)
#max_logical_replication_workers = 4    # taken from max_worker_processes
                                        # (change requires restart)
#max_sync_workers_per_subscription = 2  # taken from max_logical_replication_workers
#max_parallel_apply_workers_per_subscription = 2# taken from
max_logical_replication_workers


#------------------------------------------------------------------------------
# QUERY TUNING
#------------------------------------------------------------------------------

# - Planner Method Configuration -

#enable_async_append = on
#enable_bitmapscan = on
#enable_gathermerge = on
#enable_hashagg = on
#enable_hashjoin = on
#enable_incremental_sort = on
#enable_indexscan = on
#enable_indexonlyscan = on
#enable_material = on
#enable_memoize = on
```

```
#enable_mergejoin = on
#enable_nestloop = on
#enable_parallel_append = on
#enable_parallel_hash = on
#enable_partition_pruning = on
#enable_partitionwise_join = off
#enable_partitionwise_aggregate = off
#enable_presorted_aggregate = on
#enable_seqscan = on
#enable_sort = on
#enable_tidscan = on
#enable_group_by_reordering = on
#enable_distinct_reordering = on
#enable_self_join_elimination = on

# - Planner Cost Constants -

#seq_page_cost = 1.0                    # measured on an arbitrary scale
#random_page_cost = 4.0                 # same scale as above
#cpu_tuple_cost = 0.01                  # same scale as above
#cpu_index_tuple_cost = 0.005           # same scale as above
#cpu_operator_cost = 0.0025             # same scale as above
#parallel_setup_cost = 1000.0           # same scale as above
#parallel_tuple_cost = 0.1              # same scale as above
#min_parallel_table_scan_size = 8MB
#min_parallel_index_scan_size = 512kB
#effective_cache_size = 4GB

#jit_above_cost = 100000                # perform JIT compilation if available
                                        # and query more expensive than this;
                                        # -1 disables
#jit_inline_above_cost = 500000         # inline small functions if query is
                                        # more expensive than this; -1 disables
#jit_optimize_above_cost = 500000       # use expensive JIT optimizations if
                                        # query is more expensive than this;
                                        # -1 disables

# - Genetic Query Optimizer -

#geqo = on
#geqo_threshold = 12
#geqo_effort = 5                        # range 1-10
#geqo_pool_size = 0                     # selects default based on effort
#geqo_generations = 0                   # selects default based on effort
#geqo_selection_bias = 2.0              # range 1.5-2.0
#geqo_seed = 0.0                        # range 0.0-1.0

# - Other Planner Options -
```

```
#default_statistics_target = 100        # range 1-10000
#constraint_exclusion = partition       # on, off, or partition
#cursor_tuple_fraction = 0.1            # range 0.0-1.0
#from_collapse_limit = 8
#jit = on                               # allow JIT compilation
#join_collapse_limit = 8                # 1 disables collapsing of explicit
                                        # JOIN clauses
#plan_cache_mode = auto                         # auto, force_generic_plan or
                                        # force_custom_plan
#recursive_worktable_factor = 10.0      # range 0.001-1000000


#------------------------------------------------------------------------------
# REPORTING AND LOGGING
#------------------------------------------------------------------------------

# - Where to Log -

log_destination = 'stderr'
                                        # stderr, csvlog, jsonlog, syslog, and
                                        # eventlog, depending on platform.
                                        # csvlog and jsonlog require
                                        # logging_collector to be on.

# This is used when logging to stderr:
logging_collector = on
                                        # and csvlog into log files. Required
                                        # to be on for csvlogs and jsonlogs.
                                        # (change requires restart)

# These are only used if logging_collector is on:
#log_directory = 'log'                  # directory where log files are written,
                                        # can be absolute or relative to PGDATA
#log_filename = 'postgresql-%Y-%m-%d_%H%M%S.log'    # log file name pattern,
                                                    # can include strftime() escapes
log_file_mode = 0640                    # creation mode for log files,
                                        # begin with 0 to use octal notation
#log_rotation_age = 1d                          # Automatic rotation of logfiles will
                                        # happen after that time.  0 disables.
#log_rotation_size = 10MB               # Automatic rotation of logfiles will
                                        # happen after that much log output.
                                        # 0 disables.
#log_truncate_on_rotation = off                 # If on, an existing log file with the
                                        # same name as the new log file will be
                                        # truncated rather than appended to.
                                        # But such truncation only occurs on
                                        # time-driven rotation, not on restarts
```

```
                                        # or size-driven rotation.  Default is
                                        # off, meaning append to existing files
                                        # in all cases.

# These are relevant when logging to syslog:
#syslog_facility = 'LOCAL0'
#syslog_ident = 'postgres'
#syslog_sequence_numbers = on
#syslog_split_messages = on

# This is only relevant when logging to eventlog (Windows):
# (change requires restart)
#event_source = 'PostgreSQL'

# - When to Log -

#log_min_messages = warning             # values in order of decreasing detail:
                                        #   debug5
                                        #   debug4
                                        #   debug3
                                        #   debug2
                                        #   debug1
                                        #   info
                                        #   notice
                                        #   warning
                                        #   error
                                        #   log
                                        #   fatal
                                        #   panic

#log_min_error_statement = error        # values in order of decreasing detail:
                                        #   debug5
                                        #   debug4
                                        #   debug3
                                        #   debug2
                                        #   debug1
                                        #   info
                                        #   notice
                                        #   warning
                                        #   error
                                        #   log
                                        #   fatal
                                        #   panic (effectively off)

#log_min_duration_statement = -1        # -1 is disabled, 0 logs all statements
                                        # and their durations, > 0 logs only
                                        # statements running at least this number
                                        # of milliseconds
```

```
#log_min_duration_sample = -1           # -1 is disabled, 0 logs a sample of statements
                                        # and their durations, > 0 logs only a sample of
                                        # statements running at least this number
                                        # of milliseconds;
                                        # sample fraction is determined by
log_statement_sample_rate

#log_statement_sample_rate = 1.0   # fraction of logged statements exceeding
                                        # log_min_duration_sample to be logged;
                                        # 1.0 logs all such statements, 0.0 never logs


#log_transaction_sample_rate = 0.0 # fraction of transactions whose statements
                                        # are logged regardless of their duration; 1.0 logs all
                                        # statements from all transactions, 0.0 never logs

#log_startup_progress_interval = 10s        # Time between progress updates for
                                        # long-running startup operations.
                                        # 0 disables the feature, > 0 indicates
                                        # the interval in milliseconds.

# - What to Log -

#debug_print_parse = off
#debug_print_rewritten = off
#debug_print_plan = off
#debug_pretty_print = on
#log_autovacuum_min_duration = 10min     # log autovacuum activity;
                                        # -1 disables, 0 logs all actions and
                                        # their durations, > 0 logs only
                                        # actions running at least this number
                                        # of milliseconds.
#log_checkpoints = on
#log_connections = '' # log aspects of connection setup
        # options include receipt, authentication, authorization,
        # setup_durations, and all to log all of these aspects
#log_disconnections = off
#log_duration = off # log statement duration
#log_error_verbosity = default                  # terse, default, or verbose messages
#log_hostname = off
log_line_prefix = '%t '
                                        #   %a = application name
                                        #   %u = user name
                                        #   %d = database name
                                        #   %r = remote host and port
                                        #   %h = remote host
                                        #   %L = local address
```

```
                                        #   %b = backend type
                                        #   %p = process ID
                                        #   %P = process ID of parallel group leader
                                        #   %t = timestamp without milliseconds
                                        #   %m = timestamp with milliseconds
                                        #   %n = timestamp with milliseconds (as a Unix epoch)
                                        #   %Q = query ID (0 if none or not computed)
                                        #   %i = command tag
                                        #   %e = SQL state
                                        #   %c = session ID
                                        #   %l = session line number
                                        #   %s = session start timestamp
                                        #   %v = virtual transaction ID
                                        #   %x = transaction ID (0 if none)
                                        #   %q = stop here in non-session
                                        #       processes
                                        #   %% = '%'
                                        # e.g. '<%u%%%d> '
#log_lock_waits = off                   # log lock waits >= deadlock_timeout
#log_lock_failures = off                # log lock failures
#log_recovery_conflict_waits = off     # log standby recovery conflict waits
                                        # >= deadlock_timeout
#log_parameter_max_length = -1                  # when logging statements, limit logged
                                        # bind-parameter values to N bytes;
                                        # -1 means print in full, 0 disables
#log_parameter_max_length_on_error = 0   # when logging an error, limit logged
                                        # bind-parameter values to N bytes;
                                        # -1 means print in full, 0 disables
#log_statement = 'none'                         # none, ddl, mod, all
#log_replication_commands = off
#log_temp_files = -1                    # log temporary files equal or larger
                                        # than the specified size in kilobytes;
                                        # -1 disables, 0 logs all temp files
log_timezone = 'Africa/Nairobi'

# - Process Title -

#cluster_name = ''                      # added to process titles if nonempty
                                        # (change requires restart)
#update_process_title = off


#------------------------------------------------------------------------------
# STATISTICS
#------------------------------------------------------------------------------

# - Cumulative Query and Index Statistics -
```

```
#track_activities = on
#track_activity_query_size = 1024    # (change requires restart)
#track_counts = on
#track_cost_delay_timing = off
#track_io_timing = off
#track_wal_io_timing = off
#track_functions = none                 # none, pl, all
#stats_fetch_consistency = cache     # cache, none, snapshot


# - Monitoring -

#compute_query_id = auto
#log_statement_stats = off
#log_parser_stats = off
#log_planner_stats = off
#log_executor_stats = off


#------------------------------------------------------------------------------
# VACUUMING
#------------------------------------------------------------------------------

# - Automatic Vacuuming -

#autovacuum = on                        # Enable autovacuum subprocess?  'on'
                                        # requires track_counts to also be on.
autovacuum_worker_slots = 16            # autovacuum worker slots to allocate
                                        # (change requires restart)
#autovacuum_max_workers = 3             # max number of autovacuum subprocesses
#autovacuum_naptime = 1min              # time between autovacuum runs
#autovacuum_vacuum_threshold = 50       # min number of row updates before
                                        # vacuum
#autovacuum_vacuum_insert_threshold = 1000      # min number of row inserts
                                        # before vacuum; -1 disables insert
                                        # vacuums
#autovacuum_analyze_threshold = 50      # min number of row updates before
                                        # analyze
#autovacuum_vacuum_scale_factor = 0.2   # fraction of table size before vacuum
#autovacuum_vacuum_insert_scale_factor = 0.2     # fraction of unfrozen pages
        # before insert vacuum
#autovacuum_analyze_scale_factor = 0.1  # fraction of table size before analyze
#autovacuum_vacuum_max_threshold = 100000000    # max number of row updates
                                        # before vacuum; -1 disables max
                                        # threshold
#autovacuum_freeze_max_age = 200000000         # maximum XID age before forced
vacuum
                                        # (change requires restart)
```

```
#autovacuum_multixact_freeze_max_age = 400000000      # maximum multixact age
                                                      # before forced vacuum
                                                      # (change requires restart)
#autovacuum_vacuum_cost_delay = 2ms    # default vacuum cost delay for
                                       # autovacuum, in milliseconds;
                                       # -1 means use vacuum_cost_delay
#autovacuum_vacuum_cost_limit = -1         # default vacuum cost limit for
                                       # autovacuum, -1 means use
                                       # vacuum_cost_limit

# - Cost-Based Vacuum Delay -

#vacuum_cost_delay = 0                     # 0-100 milliseconds (0 disables)
#vacuum_cost_page_hit = 1          # 0-10000 credits
#vacuum_cost_page_miss = 2                 # 0-10000 credits
#vacuum_cost_page_dirty = 20               # 0-10000 credits
#vacuum_cost_limit = 200           # 1-10000 credits

# - Default Behavior -

#vacuum_truncate = on                      # enable truncation after vacuum

# - Freezing -

#vacuum_freeze_table_age = 150000000
#vacuum_freeze_min_age = 50000000
#vacuum_failsafe_age = 1600000000
#vacuum_multixact_freeze_table_age = 150000000
#vacuum_multixact_freeze_min_age = 5000000
#vacuum_multixact_failsafe_age = 1600000000
#vacuum_max_eager_freeze_failure_rate = 0.03 # 0 disables eager scanning

#------------------------------------------------------------------------------
# CLIENT CONNECTION DEFAULTS
#------------------------------------------------------------------------------

# - Statement Behavior -

#client_min_messages = notice              # values in order of decreasing detail:
                                       #   debug5
                                       #   debug4
                                       #   debug3
                                       #   debug2
                                       #   debug1
                                       #   log
                                       #   notice
                                       #   warning
                                       #   error
```

```
#search_path = '"$user", public'        # schema names
#row_security = on
#default_table_access_method = 'heap'
#default_tablespace = ''                # a tablespace name, '' uses the default
#default_toast_compression = 'pglz'  # 'pglz' or 'lz4'
#temp_tablespaces = ''                          # a list of tablespace names, '' uses
                                        # only default tablespace
#check_function_bodies = on
#default_transaction_isolation = 'read committed'
#default_transaction_read_only = off
#default_transaction_deferrable = off
#session_replication_role = 'origin'
#statement_timeout = 0                                  # in milliseconds, 0 is disabled
#transaction_timeout = 0                        # in milliseconds, 0 is disabled
#lock_timeout = 0                               # in milliseconds, 0 is disabled
#idle_in_transaction_session_timeout = 0    # in milliseconds, 0 is disabled
#idle_session_timeout = 0                       # in milliseconds, 0 is disabled
#bytea_output = 'hex'                   # hex, escape
#xmlbinary = 'base64'
#xmloption = 'content'
#gin_pending_list_limit = 4MB
#createrole_self_grant = ''             # set and/or inherit
#event_triggers = on

# - Locale and Formatting -

datestyle = 'iso, mdy'
#intervalstyle = 'postgres'
timezone = 'Africa/Nairobi'
#timezone_abbreviations = 'Default'  # Select the set of available time zone
                                        # abbreviations.  Currently, there are
                                        #   Default
                                        #   Australia (historical usage)
                                        #   India
                                        # You can create your own file in
                                        # share/timezonesets/.
#extra_float_digits = 1                 # min -15, max 3; any value >0 actually
                                        # selects precise output mode
#client_encoding = sql_ascii            # actually, defaults to database
                                        # encoding

# These settings are initialized by initdb, but they can be changed.
lc_messages = 'English_United States.1252' # locale for system error message
                                        # strings
lc_monetary = 'English_United States.1252' # locale for monetary formatting
lc_numeric = 'English_United States.1252' # locale for number formatting
lc_time = 'English_United States.1252'          # locale for time formatting
```

```
#icu_validation_level = warning                 # report ICU locale validation
                                        # errors at the given level

# default configuration for text search
default_text_search_config = 'pg_catalog.english'

# - Shared Library Preloading -

#local_preload_libraries = ''
#session_preload_libraries = ''
#shared_preload_libraries = ''          # (change requires restart)
#jit_provider = 'llvmjit'           # JIT library to use

# - Other Defaults -

#dynamic_library_path = '$libdir'
#extension_control_path = '$system'
#gin_fuzzy_search_limit = 0
#output_plugin_libraries = 'pgoutput, test_decoding'  # approved plugins for logical decoding


#------------------------------------------------------------------------------
# LOCK MANAGEMENT
#------------------------------------------------------------------------------

#deadlock_timeout = 1s
#max_locks_per_transaction = 64             # min 10
                                        # (change requires restart)
#max_pred_locks_per_transaction = 64        # min 10
                                        # (change requires restart)
#max_pred_locks_per_relation = -2  # negative values mean
                                        # (max_pred_locks_per_transaction
                                        #  / -max_pred_locks_per_relation) - 1
#max_pred_locks_per_page = 2                # min 0


#------------------------------------------------------------------------------
# VERSION AND PLATFORM COMPATIBILITY
#------------------------------------------------------------------------------

# - Previous PostgreSQL Versions -

#array_nulls = on
#backslash_quote = safe_encoding  # on, off, or safe_encoding
#escape_string_warning = on
#lo_compat_privileges = off
#quote_all_identifiers = off
#standard_conforming_strings = on
```

```
#synchronize_seqscans = on

# - Other Platforms and Clients -

#transform_null_equals = off
#allow_alter_system = on


#------------------------------------------------------------------------------
# ERROR HANDLING
#------------------------------------------------------------------------------

#exit_on_error = off                    # terminate session on any error?
#restart_after_crash = on               # reinitialize after backend crash?
#data_sync_retry = off                          # retry or panic on failure to fsync
                                        # data?
                                        # (change requires restart)
#recovery_init_sync_method = fsync # fsync, syncfs (Linux 5.8+)


#------------------------------------------------------------------------------
# CONFIG FILE INCLUDES
#------------------------------------------------------------------------------

# These options allow settings to be loaded from files other than the
# default postgresql.conf.  Note that these are directives, not variable
# assignments, so they can usefully be given more than once.

#include_dir = '...'                    # include files ending in '.conf' from
                                        # a directory, e.g., 'conf.d'
#include_if_exists = '...'              # include file only if it exists
#include = '...'                    # include file


#------------------------------------------------------------------------------
# CUSTOMIZED OPTIONS
#------------------------------------------------------------------------------

# Add settings for extensions here
```